ChatGPT

Yes. In fact the code gives

$$f(\infty) = 72^{-10^{24}} + \sum_{k=3}^{\infty} 2^{-k} - \sum_{k=3}^{\infty} 72^{-10^{24}k} = 0.25 + \delta,$$

with $\delta>0$ (since $72^{-10^{24}}$ is positive and the $72^{-10^{24}k}$-sum is much smaller). Hence

$$a_2 = 2f(\infty) + \tfrac{1}{2} = 1 + 2\delta > 1.$$

With $a_2>1$, the dose constraint $x_1 + a_2x_2=1$ is best met by using only the faster therapy: the minimiser is $(x_1,x_2)=(0,1/a_2)$. The infinity-norm distance from $(0.1,0.9)$ to $(0,1/a_2)$ is

$$\max(|0.1 - 0|,\ |0.9 - 1/a_2|) \ = \ \max(0.1,\ 1/a_2 - 0.9).$$

Since $1/a_2\approx1$, this distance is essentially 0.1 (indeed $1/a_2=1-2\delta+\cdots>0.9$ so $1/a_2-0.9<0.1$). Thus $|(0.1,0.9)-(0,1/a_2)|_\infty=0.1<0.15$. In conclusion, $(0.1,0.9)$ is within 0.15 of the true minimiser.
